

Antara Web Platform - One-Page App Summary

What It Is

- Project Antara Web Platform is a full-stack web app combining a React + Vite frontend with a Node.js + Express backend and PostgreSQL CMS storage.
- It powers a public mission-style site plus a built-in admin dashboard for managing posts, SEO fields, and file attachments.

Who It's For

- Primary persona: Project Antara content/admin operator managing mission logs and updates through /admin.
- Secondary audience in repo copy: students, sponsors, and technically curious visitors following the mission story.
- Formal persona document: Not found in repo.

What It Does

- Provides multi-page mission website content (home, partners, events, mini-projects, newsletter, post pages).
- Runs CMS APIs for post listing/detail, create/update/delete, and draft vs published status control.
- Supports admin login with signed Bearer tokens and protected content-management endpoints.
- Accepts attachment uploads (jpg/png/webp/pdf), image optimization, and thumbnail generation.
- Generates dynamic SEO outputs: per-page meta updates, sitemap.xml, robots.txt, and rss.xml.
- Uses caching (memory and optional Redis), rate limiting, sanitization, and security middleware.

How It Works (Architecture)

- Client: React SPA in src/App.tsx handles route state and fetches /api/posts and /api/posts/:slug.
- Server: Express app in server/index.js exposes auth, CMS CRUD, backup, and SEO feed endpoints.
- Data: PostgreSQL posts table auto-initialized on startup (including seed records and indexes).
- Storage flow: uploads saved locally under /uploads (or S3-compatible storage when UPLOAD_STORAGE=s3).
- Cache flow: request -> Express -> in-memory/Redis cache check -> PostgreSQL -> cache -> JSON response.

How To Run (Minimal)

- Install deps: npm install
- Set required env: DATABASE_URL (copy .env.example values as needed; AUTH_SECRET and ADMIN_PASSWORD should be changed).
- Start local dev (frontend + backend): npm run dev
- Open: http://localhost:5173 (frontend) and http://localhost:8787 (API).
- Note: Node/npm version requirement Not found in repo.

Evidence

- README.md, package.json scripts, server/index.js, src/App.tsx, src/main.tsx, .env.example